

Lab #3: Making a SkewT from scratch**Due: 9/30 at 11:59 PM**

For this assignment, you'll create a new Jupyter notebook titled lab3.ipynb where you'll implement the thermodynamic equations to build a SkewT-logp diagram from scratch. As noted in class, SkewTs are an essential tool for visualizing vertical profile information in meteorology, along with visualizing/conceptualizing cyclic thermodynamic processes. Your code must be sufficiently legible in organization and sufficiently commented to convey full understanding. Use markdown cells, which permit writing in LaTeX, and/or comments within code blocks to do so. All components of the assignment should be contained in the notebook. You'll be graded on the basis of completeness, correctness, clarity of your code, and written interpretations. As in prior labs, you'll save your completed notebook and push it up to your repo by the deadline and that will constitute submission.

Your final, complete SkewT must include the following elements:

1. p (hPa) as the vertical dimension, decreasing logarithmically from 1000 hPa at the bottom to 100 hPa at the top.
2. T ($^{\circ}\text{C}$) as the “ x -axis,” skewed at an angle toward the right, with values from -40°C to 50°C along the bottom axis.
3. Isotherms and *dry* and *moist* adiabats (Γ_d and Γ_m) (more below).
4. Labeled axes and clearly identified curve families and sounding profiles.
5. An actual sounding (T , T_d) from a case of your choosing. You can read in the example sounding of [MetPy](#) or explore and download a sounding from [here](#) or [here](#) or any other source you find.

Skewing. You are welcome to use the function below to shift the horizontal temperature coordinate as a function of pressure. Always pass actual pressure in hPa to this function. For the vertical coordinate, you can either plot p on a logarithmic axis or plot $\ln(p/p_0)$ on a linear axis, where $p_0 = 1000$ hPa. The function works unchanged with either approach as long as its pressure input remains p .

```
def skewT(temp, pres, skew=-20):
    """
    Skews the temperature axis for a SkewT plot.
    temp: temperature values (deg C)
    pres: pressure values (hPa)
    skew: amount of skew (default of -20)
    """
    return (temp) + skew * np.log(pres/1000.0)
```

Adiabats. For your moist adiabats, you are welcome to make use of the following equations:

$$e_s(T) = (6.112 \text{ hPa}) \exp \left[17.67 \frac{T - 273.15 \text{ K}}{T - 29.65 \text{ K}} \right],$$

$$r_s(T, p) = \epsilon \frac{e_s(T)}{p - e_s(T)}, \quad \epsilon = \frac{R_d}{R_v} \approx 0.622,$$

$$\Gamma_m \equiv - \left. \frac{dT}{dz} \right|_{\text{moist}} = \frac{g}{c_{pd}} \frac{1 + \frac{L_v r_s}{R_d T}}{1 + \frac{L_v^2 r_s}{c_{pd} R_v T^2}}.$$

Determine your own preferred approach for Γ_d . Remember that you need pressure as a vertical coordinate, so there is an added step to implementing these. The above expression for e_s is the [Bolton \(1980\) approximation](#). The formula for Γ_m assumes saturation WRT liquid water only, neglecting ice processes. We are also assuming that L_v is constant. You may use the following constant values: $R_d = 287 \text{ J kg}^{-1} \text{ K}^{-1}$, $R_v = 461 \text{ J kg}^{-1} \text{ K}^{-1}$, $c_{pd} = 1004 \text{ J kg}^{-1} \text{ K}^{-1}$, $L_v = 2.5 \times 10^6 \text{ J kg}^{-1}$. You do not need any solver packages

to implement these equations, strictly speaking, but you are welcome to use a general numerical solver if needed. Your adiabat calculations and the skewing action must otherwise be your own, explicit work (using anything provided here counts as “your own”).

Deliverables within your notebook:

1. Two proofs of concept: one for the dry adiabats and one for the moist adiabats. As researchers, we routinely need to devise demonstrations that our code conforms to some known solution or behavior by comparing against theory or a published reference. Do so in a manner that you find most convincing.
2. A complete SkewT with all five elements of the above list.
3. Responses to the following prompts in markdown cells at the end of your notebook.

Interpretation prompts:

1. Why does the moist adiabatic lapse rate, $\Gamma_m = -dT/dz$, generally increase as a saturated parcel ascends and cools?
2. At a fixed pressure, how and why does Γ_m differ between warmer and colder moist adiabats?
3. “Break” your SkewT: test a wide range of skew values and describe what happens to the plot. What does this test teach you about the value of skewing? No need to show your tests; just describe your interpretations.
4. Describe two features of the sounding you chose to plot that you find particularly interesting or useful. Refer to specific pressure levels or layers and explain the physical significance of each feature.
5. What was the most challenging step for this lab? How did you navigate it, and what was the solution? If you leaned on GenAI tools, how did they contribute to your work and how did you check the information they provided?